**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# LESSON 006 · Sound, Sampling, Bit Depth and File Sizes

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. What is image metadata?

Q2. A has code 65. Capital-letter codes go in order. What is D's code?
A. 65
B. 68
C. 67
D. 69

Q3. Why can Unicode support more writing systems than standard ASCII?

Q4. A character code uses exactly 6 bits. How many different characters can it represent? Can it encode 65 different characters without more bits?

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# I Do · Read new knowledge

### Key vocabulary

analogue — A continuously varying signal.

sample rate — Number of samples taken per second.

bit depth — Number of bits representing each sample.

duration — Length of the sound recording in seconds.

### New learning

Analogue versus digital: Real sound is continuous. To digitise it, a computer samples the wave at regular intervals and stores measurements as binary.

Sample rate: Sample rate is the number of samples measured per second, in hertz (Hz). More samples can capture more detail.

Bit depth: Bit depth is the number of bits available for each sample's amplitude; more bits allow finer amplitude representation.

Raw mono sound: Raw bits = sample rate × seconds × bit depth. Divide by eight for bytes. Channels, headers and compression can change real file size.

### Worked example

A mono clip uses 10 samples per second, lasts 5 seconds and uses 8 bits per sample. Find the raw size.

1. Write the formula: rate × duration × bit depth.

2. 10 samples/s × 5 seconds = 50 samples.

3. 50 × 8 = 400 bits.

4. 400 ÷ 8 = 50 bytes (raw, mono, uncompressed).

# We Do · Practise with guidance

### Guided 1

A sound clip has sample rate 20 Hz and duration 10 s. How many samples?

Worked method: 20 × 10 = 200 samples.

### Guided 2

The clip uses 8 bits per sample. How many raw bits?

Worked method: 200 × 8 = 1,600 bits.

### Guided 3





**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

## Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. Extra image details, such as width, height or colour depth. These are separate from pixel data.

Q2. B — 68

Q3. Unicode supports far more characters than standard ASCII.

Q4. (a) 2^6 = 64; no, one more character requires more distinct codes.

### I Do / We Do pages · Guided practice

Q1. 20 × 10 = 200 samples.

Q2. 200 × 8 = 1,600 bits.

Q3. 1,600 ÷ 8 = 200 bytes; doubling duration doubles the data size to 400 bytes.

### You Do pages · Code First

```
rate = 20
seconds = 4
depth = 8
print(rate * seconds * depth)
```

Expected result: 640

### You Do pages · Independent questions

Q1. The number of sound samples taken [1] every second [1].

Q2. More bits represent each sample [1], allowing more accurate representation of amplitude [1], but increasing raw file size [1].

Q3. 1,000 × 4 = 4,000 samples [1]; ×8 = 32,000 bits [1]; ÷8 = 4,000 bytes [1].

### Exit Ticket · Model answer

True

## Try the question again

Choose one question you got wrong. All correct? Choose the hardest.



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BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

Calculate bytes and state how the size changes if duration doubles.

Worked method: 1,600 ÷ 8 = 200 bytes; doubling duration doubles the data size to 400 bytes.

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

## You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Code First · Predict → Write → Test → Improve

Predict the sound size in bits.

```
rate = 20
seconds = 4
depth = 8
print(rate * seconds * depth)
```

predict

### Question 1

State what is meant by sample rate. [2 marks]

### Question 2

Explain how increasing bit depth affects sound quality and raw file size. [3 marks]

### Question 3

A mono recording is sampled at 1,000 Hz for 4 seconds, using 8 bits per sample. Calculate its raw size in bytes. [3 marks]

## Exit Ticket · Answer before checking





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Doubling sample rate doubles uncompressed mono sound data size when all other values stay fixed.